INTRODUCTION

This lab focuses on correlation analysis and normalization techniques in R. Correlation analysis is used to identify relationships between variables, while normalization techniques are applied to transform data into comparable scales. These methods are important in data analysis and preprocessing for better interpretation and modelling.

ACTIVITY 1: CORRELATION ANALYSIS USING ToothGrowth DATASET

1.1 DATASET DESCRIPTION

The **ToothGrowth** dataset contains information on tooth length (len), supplement type (supp), and dosage (dose). In this analysis, only the numeric variables (len and dose) are used for correlation analysis.

1.2 CORRELATION ANALYSIS

The Pearson correlation coefficient between len and dose was computed using cor() and further validated using cor.test(). The correlation results obtained from the analysis are shown below.

```
> cor(ToothGrowth[, c("len", "dose")])
           len      dose
len  1.0000000 0.8026913
dose 0.8026913 1.0000000
> cor.test(ToothGrowth$len, ToothGrowth$dose, method = "pearson")

        Pearson's product-moment correlation

data:  ToothGrowth$len and ToothGrowth$dose
t = 10.25, df = 58, p-value = 1.233e-14
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.6892521 0.8777169
sample estimates:
      cor
0.8026913
```

Figure 1.1: Pearson Correlation Analysis Between Tooth Length and Dose

1.3 DATA VISUALIZATION AND INSIGHTS

A correlation heatmap was generated to visually represent the relationship between variables.

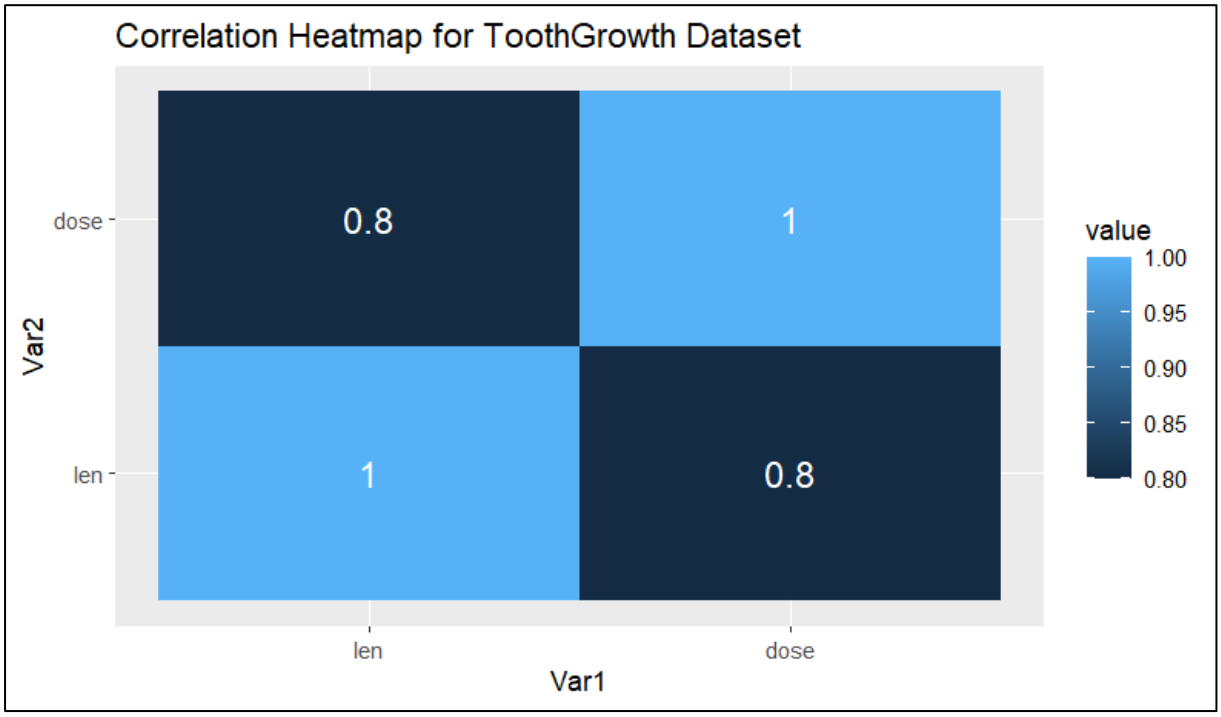

Figure 1.2: Correlation Heatmap for ToothGrowth Dataset

Observation:

The correlation coefficient between len and dose is approximately **0.80**, indicating a strong positive relationship.

This means that as the dosage increases, the tooth length also increases.

Insight:

The correlation is statistically significant (p-value < 0.05), suggesting that the relationship is not due to random chance.

This indicates that dosage plays an important role in tooth growth.

Conclusion:

The heatmap clearly highlights the strong relationship between the variables, demonstrating the usefulness of correlation analysis in identifying patterns.

ACTIVITY 2: NORMALIZATION USING mtcars DATASET

2.1 DATASET DESCRIPTION

The **mtcars** dataset contains information about various car attributes. In this activity, the mpg variable is used to demonstrate normalization techniques.

2.2 NORMALIZATION METHODS

2.2.1 Log Transformation

The data was transformed using the logarithmic function, which compresses large values and reduces skewness. The result of the log transformation is shown below.

```
> scaled_data1 <- log(mydata)
> print(scaled_data1)
 [1] 3.044522 3.044522 3.126761 3.063391 2.928524 2.895912 2.660260 3.194583 3.126761 2.954910
[11] 2.879198 2.797281 2.850707 2.721295 2.341806 2.341806 2.687847 3.478158 3.414443 3.523415
[21] 3.068053 2.740840 2.721295 2.587764 2.954910 3.306887 3.258097 3.414443 2.760010 2.980619
[31] 2.708050 3.063391
>
```

Figure 2.1: Log transformation of MPG Values

2.2.2 Standard Scaling

The data was scaled to have a mean of 0 and a standard deviation of 1, allowing comparison relative to the mean. The result of standard scaling is shown below.

```
> scaled_data2 <- as.data.frame(scale(mydata))
> print(scaled_data2)
            V1
1   0.15088482
2   0.15088482
3   0.44954345
4   0.21725341
5  -0.23073453
6  -0.33028740
7  -0.96078893
8   0.71501778
9   0.44954345
10 -0.14777380
11 -0.38006384
12 -0.61235388
13 -0.46302456
14 -0.81145962
15 -1.60788262
16 -1.60788262
17 -0.89442035
18  2.04238943
19  1.71054652
20  2.29127162
21  0.23384555
22 -0.76168319
23 -0.81145962
24 -1.12671039
25 -0.14777380
26  1.19619000
27  0.98049211
28  1.71054652
29 -0.71190675
30 -0.06481307
31 -0.84464392
32  0.21725341
>
```

Figure 2.2: Standard Scaling of MPG Values (Z-score Normalization)

2.2.3 Min-Max Scaling

The data was scaled between 0 and 1, making it easier to compare values within a fixed range. The result of min-max scaling is shown below.

```
> minmax <- preProcess(as.data.frame(mydata), method = c("range"))
> scaled_data3 <- predict(minmax, as.data.frame(mydata))
>
> print(scaled_data3)
      mydata
1  0.4510638
2  0.4510638
3  0.5276596
4  0.4680851
5  0.3531915
6  0.3276596
7  0.1659574
8  0.5957447
9  0.5276596
10 0.3744681
11 0.3148936
12 0.2553191
13 0.2936170
14 0.2042553
15 0.0000000
16 0.0000000
17 0.1829787
18 0.9361702
19 0.8510638
20 1.0000000
21 0.4723404
22 0.2170213
23 0.2042553
24 0.1234043
25 0.3744681
26 0.7191489
27 0.6638298
28 0.8510638
29 0.2297872
30 0.3957447
31 0.1957447
32 0.4680851
>
```

Figure 2.3: Min-Max Scaling of MPG Values (Range 0 to 1)

Observation:

Each normalization method transforms the data differently, resulting in distinct value distributions:

- Log transformation compresses values
- Standard scaling centers the data around zero
- Min-max scaling restricts values between 0 and 1

Comparison:

- Log Transformation → reduces skewness

- Standard Scaling → standardizes variation
- Min-Max Scaling → normalizes range

Insight:

Different normalization techniques serve different purposes, and the choice depends on the type of analysis being performed.

Conclusion:

Normalization improves data comparability and prepares it for further analysis. Understanding the differences between methods is essential in data preprocessing.

3.0 OVERALL CONCLUSION

This lab demonstrates how correlation analysis and normalization techniques help in understanding relationships between variables and preparing data for analysis. These methods are essential tools in data preprocessing and play a crucial role in effective data analysis and modelling.